CREDIT CARD DEFAULT PREDICTION USING MACHINE LEARNING

1. PROJECT OVERVIEW

The Credit Card Default Prediction project aims to predict whether a credit card customer will default on their next payment using machine learning techniques. Financial institutions face significant risks due to customer loan defaults, and predicting such behavior helps banks reduce financial losses and improve risk management strategies.

This project applies machine learning algorithms to analyze customer demographic information, payment history, bill amounts, and previous payment records to determine the likelihood of default.

2. PROBLEM STATEMENT

Banks and financial institutions provide credit cards to millions of customers. However, some customers fail to repay their credit card bills on time, resulting in financial losses. Identifying potential defaulters before issuing or extending credit is crucial for effective risk management.

The objective of this project is to build a machine learning model that predicts whether a customer will default on their next credit card payment based on historical customer data.

3. DATASET INFORMATION

The dataset used in this project is the Credit Card Default Dataset, which contains customer demographic information, credit details, and repayment history.

Dataset Statistics:

- Total Records: 30,000
- Total Features: 24
- Target Variable: Default Payment Next Month
- Problem Type: Binary Classification

Target Variable:

- 0 = No Default
- 1 = Default

4. TECHNOLOGIES USED

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Development Environment

- Jupyter Notebook
- Google Colab

Dataset Format - CSV File

5. MODELS USED

The machine learning model used in this project is:

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of a customer defaulting based on historical financial information.

Other models that can also be used:

- Decision Tree
- Random Forest
- Support Vector Machine
- XGBoost

Reasons for choosing Logistic Regression:

- Easy to implement
- Fast training process
- Good performance for classification problems
- Easy interpretation of results

6. EVALUATION METRICS

The model performance is evaluated using the following metrics:

Accuracy Score

Measures the percentage of correctly predicted observations.

Confusion Matrix

Displays:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

precision

Measures the proportion of correctly predicted positive observations.

Recall

Measures the ability of the model to identify actual defaulters.

F1-Score

Provides a balance between precision and recall.

ROC-AUC Score

Measures the overall classification performance of the model.

7. RESULTS

After training and testing the model:

- Training Data: 80%
- Testing Data: 20%

Model Performance:

- Accuracy: Approximately 81% to 83%
- Precision: Good
- Recall: Moderate
- F1-Score: Balanced
- ROC-AUC Score: Satisfactory

The model successfully identifies customers who are likely to default on their credit card payments, helping financial institutions reduce credit risk.

8. REPOSITORY STRUCTURE

```
Credit-Card-Default-Prediction/
│
├── dataset/
│   └── credit_card_default.csv
│
├── notebooks/
│   └── Credit_Card_Default_Prediction.ipynb
│
├── models/
│   └── logistic_model.pkl
│
├── reports/
│   └── Project_Report.docx
│
├── images/
│   └── confusion_matrix.png
│
├── README.md
│
└── requirements.txt
```

9. FUTURE ENHANCEMENTS

The project can be further improved by:

- Using advanced algorithms such as XGBoost and Random Forest.
- Performing feature engineering and feature selection.
- Applying hyperparameter tuning.
- Handling class imbalance using SMOTE.
- Developing a real-time prediction dashboard.
- Deploying the model using Flask or Streamlit.
- Integrating the model with banking applications.

10. CONCLUSION

This project successfully demonstrates the use of machine learning for predicting credit card payment defaults. The Logistic Regression model provides satisfactory prediction accuracy and can assist financial institutions in making informed credit decisions. Early identification of potential defaulters helps reduce financial risk and improve customer credit management. Future improvements and advanced algorithms can further enhance the model's prediction performance and practical usability.